

SCTR's Pune Institute of Computer Technology, Dhankawadi, Pune

A PROJECT REPORT ON

Design and Development of ESDM Virtual Lab - Role-Based Learning Platform

SUBMITTED BY

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CERTIFICATE

This is to certify that the curriculum-based project report entitled **Design and Development of ESDM Virtual Lab - Role-Based Learning Platform**, submitted by **Vivek Jitendra Patil** (Roll No: 33347), has been satisfactorily completed under the guidance of **Mrs. Swapnaja R. Hiray** towards the partial fulfillment of Third Year Information Technology, Semester VI, Academic Year 2025-26 of Savitribai Phule Pune University.

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I also thank the faculty of the Department of Information Technology, PICT, for providing a strong academic foundation and the opportunity to execute this role-based, full-stack application as part of curriculum-based learning.

This project helped me strengthen my practical understanding of backend API design, mobile app development, role-based access, cloud integrations, and real-world deployment considerations.

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1. Introduction

The ESDM Virtual Lab project was designed as a role-based digital learning platform to support teaching, assessment, and content distribution for Electronics System Design and Manufacturing (ESDM) coursework. The platform combines a Node.js and Express backend with a MongoDB data layer and an Expo React Native frontend that serves both Teacher and Student roles through dedicated interfaces.

The project architecture was planned to cover the complete academic workflow: secure authentication, class-wise assignment publishing, quiz creation through Excel upload, quiz activation and attempt tracking, digital notes and video resources, diagram sharing, and profile management.

Compared with conventional classroom workflows where resources are scattered across messaging apps, emails, and manual records, this virtual lab provides a unified and trackable system with clear role boundaries and centralised data access.

2. Problem Statement

Traditional lab and coursework management in many institutes depends on fragmented tools and manual processes. This creates delays, data inconsistency, and weak visibility into student progress. For ESDM subjects specifically, students need access to assignments, notes, diagram references, and quizzes in a structured manner, while teachers need streamlined content publishing and monitoring capabilities.

- No single platform to manage assignments, quizzes, diagrams, notes, and video links together.
- Difficulty in enforcing role-based access between students and teachers.
- Manual quiz workflows without bulk question upload and activation control.
- Lack of class/batch-based filtering for educational resources.
- No integrated dashboard view for engagement and performance indicators.
- High operational overhead while tracking profile updates and learning artifacts.

The ESDM Virtual Lab addresses these challenges by introducing a secure, API-driven, role-aware learning ecosystem that supports both mobile usability and backend extensibility.

3. Objectives and Scope

3.1 Objectives

1. To build a role-based educational platform for Student, Teacher, and Admin-enabled architecture.
2. To implement secure authentication and protected API access using JWT.
3. To provide teacher workflows for creating assignments, quizzes, notes, diagrams, and video resources.
4. To provide student workflows for viewing resources and submitting quiz responses.
5. To support Excel-based quiz ingestion to reduce manual content entry effort.
6. To expose dashboard endpoints for analytical summaries of learning activity.
7. To integrate cloud media storage and file workflows for scalable content handling.

3.2 Scope

The scope includes backend API implementation, MongoDB schema design, role-based middleware, and a complete mobile frontend for teacher and student operations. Covered modules include authentication, profile management, assignment workflows, quiz lifecycle (creation, activation, attempt, submission), notes, diagrams, videos, student analytics, and dashboard metrics.

The scope excludes enterprise-scale load testing, advanced AI recommendation systems, and comprehensive offline synchronization.

4. Methodological Details

4.1 Designing and Developing Role-Based E-Learning Platform

The project followed an iterative development model. Each module was implemented in phases: data model design, API development, middleware integration, frontend screen implementation, end-to-end testing, and refinement.

4.1.1 Backend Architecture (Node.js, Express, MongoDB)

- Express server with modular routes for auth, profile, assignments, quizzes, notes, videos, diagrams, students, and dashboards.
- MongoDB models designed for users, assignments, quizzes, quiz attempts, notes, video links, and diagrams.
- JWT-protected middleware to secure role-restricted endpoints.
- Cloudinary integration for media and document handling.
- Excel parsing in quiz module for bulk question import.

4.1.2 Student Module (Expo React Native)

- Dedicated student drawer navigation with dashboard, assignments, quizzes, notes, diagrams, videos, and profile views.
- Access to active quizzes with submit flow and score-oriented result handling.
- Batch-filtered access to learning resources published by teachers.

4.1.3 Teacher Module (Expo React Native)

- Teacher dashboard with key academic indicators and quick-access actions.
- Content creation workflows for assignments, notes, quizzes, videos, and diagrams.
- Quiz activation and status control to govern student availability windows.
- Student listing and profile/analytics views for performance observation.

4.1.4 API Service and Frontend Integration

Axios-based service modules were created for assignment, quiz, dashboard, student, video, and diagram APIs. Authentication tokens were persisted in AsyncStorage and attached to protected requests. This modular service layer reduced coupling between UI screens and transport logic, improving maintainability.

4.2 Deployment and Integration Workflow

The frontend and backend were designed for independent deployment with clear API contracts. The backend can run via Node.js process management and connect to MongoDB Atlas, while the frontend uses Expo tooling for Android, iOS, and web previews. Deployment readiness included environment variable management, CORS configuration, and endpoint validation using Postman-style testing flows.

5. Modern Engineering Tools Used

Category	Tool / Technology	Purpose
Frontend Framework	Expo Router (React Native)	Mobile application with role-based navigation
Backend Runtime	Node.js + Express	REST API implementation
Database	MongoDB + Mongoose	Persistent storage and schema modeling
Authentication	JWT	Secure role-based endpoint protection
Cloud Media	Cloudinary + Multer	File upload and media hosting
Spreadsheet Parsing	xlsx	Quiz question import through Excel files
API Client	Axios	Frontend to backend communication
State Persistence	AsyncStorage	Token and session data storage
Charts	react-native-chart-kit	Dashboard visualization
Version Control	Git & GitHub	Source management and collaboration
IDE / Editor	VS Code	Development and debugging
API Testing	Postman	Endpoint verification

The project stack was chosen to balance rapid iteration, modular architecture, and deployment flexibility for an academic environment.

6. Outcome / Result of Project Work

6.1 Technical Outcomes

- Developed and integrated a complete role-based learning platform for ESDM coursework.
- Implemented secure login and protected API access paths for multiple user roles.
- Enabled teacher-driven academic content publishing and class-targeted distribution.
- Implemented quiz lifecycle with Excel import, activation control, and student submission tracking.
- Delivered dashboard and analytics endpoints for performance visibility.
- Integrated cloud workflows and modular API services for maintainable growth.

6.2 Module-Wise Summary

Module	Primary Users	Key Deliverables
Authentication & Profile	Student, Teacher	Login, register, profile update, password flow
Assignments	Teacher, Student	Create/view assignments, class-wise publishing
Quizzes	Teacher, Student	Excel upload, activation, attempt, submit, score tracking
Notes / Videos / Diagrams	Teacher, Student	Resource management with batch-based visibility
Students & Dashboard	Teacher	Student listing, analytics and overview metrics

6.3 Learning Outcomes

- Strengthened practical understanding of full-stack architecture across backend APIs and mobile clients.
- Gained experience in role-based data access and middleware-oriented security design.
- Improved skill in schema modeling for educational workflows and entity relationships.
- Practiced modular API service abstraction and multi-screen navigation in Expo Router.
- Developed stronger debugging, integration testing, and deployment-readiness practices.

6.4 Screenshots of Project Outputs

The following pages are reserved for screenshots of key interfaces such as login, teacher dashboard, student home, quiz flow, and analytics modules.

Project Screenshots - Authentication and Navigation

<i>Insert screenshots here</i>

Project Screenshots - Teacher Modules

<i>Insert screenshots here</i>

Project Screenshots - Student Modules

<i>Insert screenshots here</i>

Project Screenshots - Dashboards and Analytics

<i>Insert screenshots here</i>

- End of Project Report -